NAS Solution

Rockstor NAS Implementation

Built on openSUSE · Powered by BTRFS

CLOIF2 PROJECT · Yona Jervis & Donovan Glodt · LGK – 2026

AGENDA

Rockstor

01 **What is Rockstor?**
NAS software overview & origins

02 **Core Features**
BTRFS, snapshots, RAID, protocols

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Share layout, users, services

04 **Protocols & iSCSI**
SMB, NFS, SFTP, iSCSI deep-dive

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Honest assessment & comparison

What is Rockstor?

NAS Operating System

Rockstor is an open-source Linux-based NAS operating system built on openSUSE Leap, using the BTRFS filesystem as its core.

It provides a browser-based Web UI, advanced storage management, and Cloud Storage features out of the box.

Supported architectures: x86_64, ARM64EFI, Raspberry Pi 4.

Fully browser-managed. No CLI knowledge needed for day-to-day use; pools, shares, users and services are all handled through the Web UI

Founded by

Suman Chakravartula
(Open Collective Non-Profit)

Maintained by

Philip Paul Guyton & Flox

Latest Release

Rockstor 5.1.0-0

Pricing

Free (Testing) · 24€/yr (Stable)

Core Features · BTRFS Foundation

Rockstor

Copy-on-Write (CoW)

Data is never overwritten — all changes create new copies. Enables instant snapshots with near-zero overhead.

Bitrot Protection

Checksums on all data blocks detect and flag silent corruption automatically.

Dynamic Pool Scaling

Add or remove drives of any size on the fly without interrupting services.

Online RAID Changes

Switch between RAID profiles (RAID1, RAID5, RAID6...) while the system is running.

Instant Snapshots

Create manual or scheduled point-in-time snapshots. Roll back from mistakes in seconds.

Replication

Appliance-to-appliance snapshot replication for off-site disaster recovery.

Core Features · Sharing & Management

Rockstor

SMB / CIFS	NFS	SFTP	AFP

Web Dashboard

Full administration through a clean browser-based UI — no CLI required for day-to-day operations.

User & Group Management

Create users and groups, assign per-share permissions with granular access control.

Active Directory / LDAP

Seamless enterprise integration for centralised authentication.

Rock-ons (Docker)

One-click application deployment: Plex, Nextcloud, Syncthing and more.

Scheduled Tasks

Automate scrubs, snapshots, and replication on custom schedules.

Email Alerts

Real-time notifications for hardware events, disk failures and pool warnings.

Implementation · Overview

Rockstor

Scenario: Family NAS with private spaces, shared photo library, and network-attached virtual disks

Software RAID 1

2× 160 GB SATA HDDs
(Mainboard)

Hardware RAID 1

2× 156 GB SAS HDDs
(RAID Controller)

Boot SSD

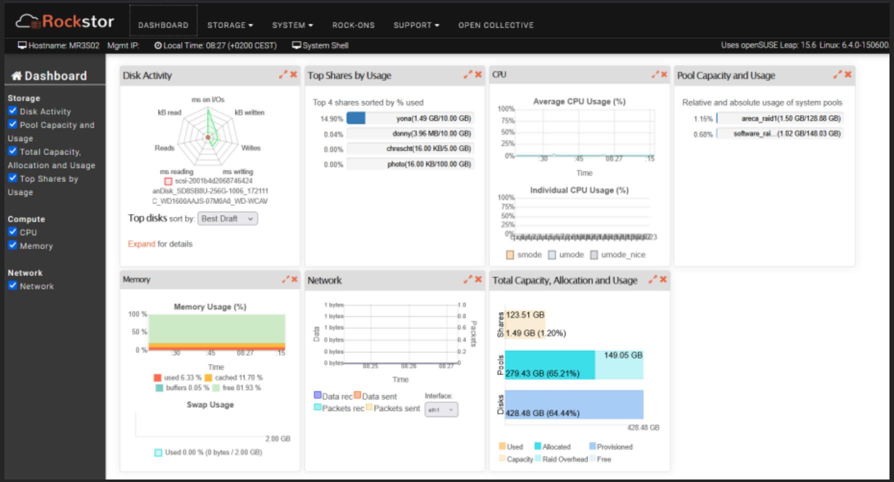
1× 256 GB SSD
Rockstor OS

Share Layout

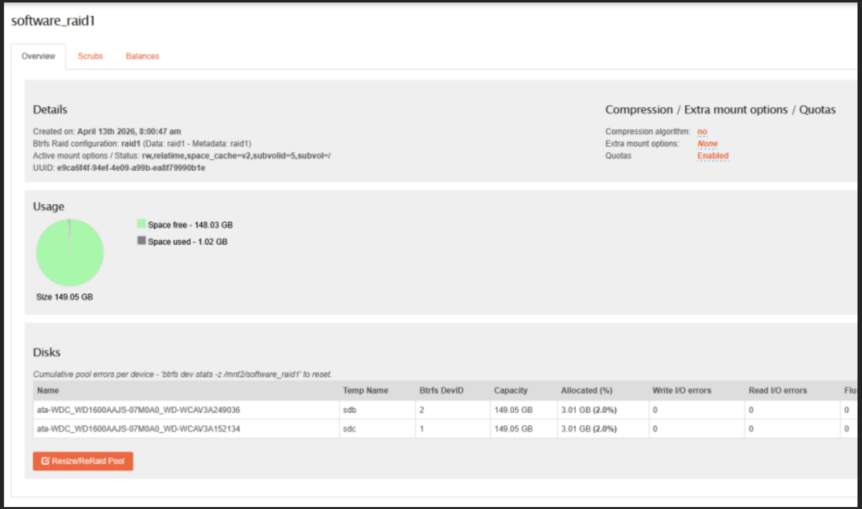
Share	Size	Storage	Access
yona	10 GB	Software RAID	Owner only
donny	10 GB	Software RAID	Owner only
chrëscht	5 GB	Software RAID	Owner only
photo	~100 GB	Hardware RAID	All users Read & Write
iSCSI	1 GB	Software RAID	Virtual disk

Implementation · Screenshots

Rockstor Dashboard & Pool Creation



Dashboard



Software RAID

Implementation · Users, Groups & Shares

Users

yona

Group: family

Linux + Windows + SFTP

donny

Group: family

Windows (SMB)

chrëscht

Group: family

Android (SFTP)

Username
chrescht
donny
yona

Users

Name	Size	Usage
chrescht	5.00 GB	16.00 KB
donny	10.00 GB	3.96 MB
photo	100.00 GB	16.00 KB
yona	10.00 GB	1.49 GB

Shares

Protocols & Access Matrix

Rockstor

SMB / Samba

Windows

File sharing for Windows clients. Private shares (yona, donny) exported via SMB with Workgroup 'MYGROUP'.

NFS

Linux

Unix/Linux native protocol using Remote Procedure Calls. Yona accesses their share from a Linux desktop over NFS.

SFTP

Android / Cross-platform

Secure file transfer over SSH. Chrëscht accesses their 5 GB share from Android and mobile devices via SFTP.

iSCSI

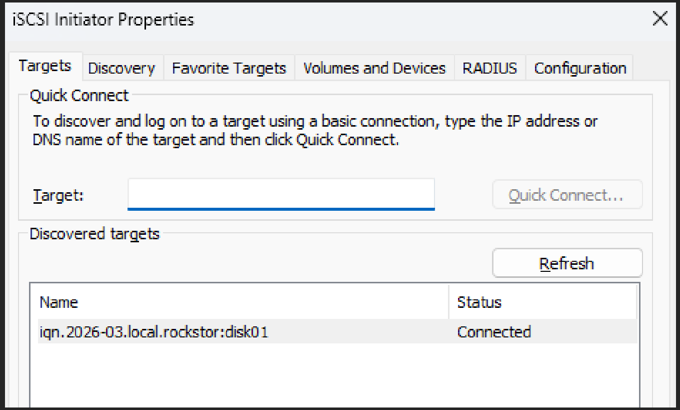
Virtual Disk

Block-level storage over TCP/IP — the drive appears as a native disk. Configured via targetcli on the software RAID.

iSCSI · Block Storage over the Network

Setup

- iSCSI isn't natively supported by Rockstor
- Installation must be done through targetCLI through the servers CLI
- Once configured, the iSCSI virtual disk shows up in Windows Disk Management exactly like a physical local drive.



iSCSI on local machine

Demonstration

Strengths · Why Rockstor Works

BTRFS-Native Architecture

Built from the ground up on BTRFS. Snapshots, CoW, checksums and dynamic RAID are first-class citizens.

Clean Web UI

Manage everything through a browser. No need to memorise commands needed for daily operations.

Free Tier Available

Full feature set at zero cost. The free 'Testing Updates' tier is solid for small projects.

Rock-ons (Docker Apps)

Allows for deployment of Nextcloud, Plex, Syncthing on the server.

Hardware Flexibility

Runs on x86, ARM64EFI, Raspberry Pi 4.

Appliance Replication

Native snapshot-based replication between Rockstor instances for simple off-site backup.

Limitations

Smaller Community & Ecosystem

Fewer forum posts, tutorials and plugins than TrueNAS, OpenMediaVault or Unraid.

iSCSI Not Natively Supported

No Web UI panel for iSCSI — requires CLI setup via targetcli.

Paid Tier for Stable Updates

Stable tested updates cost 24€/year. Free tier runs 'testing' packages.

BTRFS Maturity

BTRFS is younger than ZFS. RAID 5/6 levels are still considered experimental for production data.

No Built-in VM Support

No native hypervisor. Running VMs requires Docker workarounds via Rock-ons.

Licenses

FREE · Testing Tier

€0 / year

- ✓ Full feature set
- ✓ BTRFS pools, shares, snapshots
- ✓ All protocols (SMB, NFS, SFTP…)
- ✓ Rock-ons (Docker apps)
- ✓ Testing updates
- ✓ Community support (forum)

STABLE · Recommended

€24 / year

via opencollective.com

- ✓ Everything in Free
- ✓ Stable, tested updates
- ✓ High-priority bug fixes
- ✓ Paid support available
- ✓ Supports the open-source project

Summary

Rockstor

1

Modern, BTRFS-first NAS

The most BTRFS-integrated NAS OS with snapshots, live RAID, data integrity, all built-in.

2

Accessible & Affordable

Free tier covers everything. 24€/year for stable updates.

3

Ideal for SMBs

Easy enough for home labs, powerful enough for small business. Runs on hardware you already own.